

Probability
Chapter 4 - Module 3

Hardy-Weinberg Law

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We study a single individual gene and suppose that the gene can be one of two types, say A or a .

Assuming that each individual has two genes, then they can be of type AA , Aa or aa .

Let the initial fraction of each gene pair, AA , aa and Aa be p_0 , q_0 and r_0 respectively, where $p_0 + q_0 + r_0 = 1$, so that these are the only possible cases.

We assume that mating is random in the following sense - the probability is equally likely that any individual will mate with any other individual.

Question: What is the distribution of gene pairs in the next generation?

Let's compute the probability that a random gene is A or a.

We will denote these at $P(A)$ and $P(a)$

We will need some conditional probabilities.

So let's write $P(A | AA)$ to denote the probability that of A is passed to the offspring given that the mother has type AA.

Similarly write $P(A | Aa)$ to denote the probability that of A is passed to the offspring given that the mother has type Aa.

Etc.

First of all we note that using conditional probabilities that

$$\begin{aligned}P(A) &= P(A|AA)p_0 + P(A|aa)q_0 + P(A|Aa)r_0 \\&= 1 \cdot p_0 + 0 \cdot q_0 + 1/2 \cdot r_0 \\&= p_0 + r_0/2.\end{aligned}$$

Similarly

$$\begin{aligned}P(a) &= P(a|AA)p_0 + P(a|aa)q_0 + P(a|Aa)r_0 \\&= 0 \cdot p_0 + 1 \cdot q_0 + 1/2 \cdot r_0 \\&= q_0 + r_0/2.\end{aligned}$$

So if we let p , q and r be the new probabilities of gene pairs AA, aa, and Aa, then because each individual passes one random gene to its offspring we have

$$\begin{aligned}p &= P(A)P(A) = (p_0 + r_0/2)^2 \\q &= P(a)P(a) = (q_0 + r_0/2)^2\end{aligned}$$

and

$$r = 2P(A)P(a) = 2(p_0 + r_0/2)(q_0 + r_0/2).$$

So what are the proportion of A and a in this second generation. It is not too surprising that it has not changed.

We can see this by noting that the probability that a gene is A in the second generation is

$$\begin{aligned} p + r/2 &= (p_0 + r_0/2)^2 + 2(p_0 + r_0/2)(q_0 + r_0/2)/2 \\ &= (p_0 + r_0/2)[(p_0 + r_0/2) + (q_0 + r_0/2)] \\ &= (p_0 + r_0/2)[(p_0 + q_0 + r_0)] \\ &= (p_0 + r_0/2) \\ &= P(A) \end{aligned}$$

since $p_0 + q_0 + r_0 = 1$.

And the probability that a gene is a in the second generation is

$$1 - (p + r/2) = 1 - P(A) = P(a).$$

What is less obvious is that the proportion of pairs AA, aa and Aa are constant after the first generation. This is known as the Hardy-Weinberg law. This can be established in several different ways.

Let's define a chain X_n with states AA, aa and Aa. Given the initial probabilities p_0 , q_0 and r_0 , let

$$p = p_0 + r_0/2$$

$$q = q_0 + r_0/2$$

and

$$r = 1 - p - q.$$

What are the transition probabilities to the next step in the chain?

Suppose that mom is Aa. What is the probability that the Child is AA, aa and Aa?

Let C denote child.

Let F denote father.

Then since Mom is Aa,

$$\begin{aligned} P(C \text{ is AA}) &= P(C \text{ is AA} | F \text{ passes A}) P(F \text{ passes A}) \\ &+ P(C \text{ is AA} | F \text{ passes a}) P(F \text{ gives a}) \\ &= \frac{1}{2} (p + r/2) + 0 (q + r/2) \\ &= p/2 + r/4. \end{aligned}$$

$$\begin{aligned}
P(\text{C is Aa}) &= P(\text{C is Aa} | \text{F passes A}) P(\text{F passes A}) \\
&+ P(\text{C is Aa} | \text{F passes a}) P(\text{F gives a}) \\
&= \frac{1}{2} (p + r/2) + \frac{1}{2} (q + r/2) \\
&= p/2 + q/2 + r/2.
\end{aligned}$$

And finally

$$\begin{aligned}
P(\text{C is aa}) &= P(\text{C is aa} | \text{F passes A}) P(\text{F passes A}) \\
&+ P(\text{C is aa} | \text{F passes a}) P(\text{F gives a}) \\
&= 0 (p + r/2) + \frac{1}{2} (q + r/2) \\
&= q/2 + r/4.
\end{aligned}$$

So here is the transition probability matrix for this simple genetics problems.

$$\begin{array}{c}
 \begin{array}{ccc}
 & AA & aa & Aa \\
 \begin{array}{c}
 AA \\
 aa \\
 Aa
 \end{array}
 & \begin{pmatrix}
 p + \frac{r}{2} & 0 & q + \frac{r}{2} \\
 0 & q + \frac{r}{2} & p + \frac{r}{2} \\
 \frac{p}{2} + \frac{r}{4} & \frac{q}{2} + \frac{r}{4} & \frac{p}{2} + \frac{q}{2} + \frac{r}{2}
 \end{pmatrix}
 \end{array}
 \end{array}$$